

GLOSSARY

4x4 – A power-endurance training exercise in which four (or any number) boulder problems are climbed in quick succession, followed by a rest period, then repeated until a total of four (or any number) sets has been completed.

Actin – Thin filament proteins within the muscle's Sarcomeres that myosin proteins bind to in order to contract the muscle.

Action Potential – An impulse from the central nervous system that initiates muscle contractions.

Active Rest – Non-climbing exercises performed during the Rest Phase that promote general health and well-being.

Acute Injury – An injury resulting from a specific event or trauma, as opposed to an Overuse Injury.

Adaptation – The body's process of growth and change in order to improve physical, technical or mental abilities.

Aerobic Respiration – A very efficient, but slow, energy metabolism process that requires oxygen.

Anaerobic – A less-efficient energy metabolism process that does not require oxygen.

Aerobic Restoration and Capillarity (ARC) – A form of base-fitness training consisting of continuous climbing for 20 - 45 minutes, right at the climber's Maximum Steady State level of intensity.

Arousal – A psychological state of alertness and readiness to react to challenging situations.

Adenosine Di-Phosphate (ADP) – A byproduct of ATP Hydrolysis during muscle energy metabolism that contributes to fatigue.

Adenosine Tri-Phosphate (ATP) – The primary fuel for many bio-chemical functions in the body, especially muscle contractions.

ATP Hydrolysis (aka ATPase) – The chemical reaction that converts ATP and water into ADP, P_i , and H^+ , while releasing energy for cellular functions, such as muscle cell contractions.

Base-Fitness – The body's capacity to fuel muscle contractions for sustained activity.

Baseline – An athlete's standard ability at the beginning of any training season, phase, or exercise. The baseline changes as the athlete improves or de-trains over the course of a career.

Calcium Ions (Ca^{2+}) – Minerals in the muscle cells that trigger cross bridge cycling by the actin and myosin (muscle contractions) when released from the Sarcoplasmic Reticulum.

Campus Board – A ladder of finger edges (typically wood) fixed to a slightly overhanging wall.

Campus – A climbing movement that is performed without the feet touching the rock. It requires power and momentum to move the hands dynamically from one hold (or wooden rung on a campus board) to the next while the feet hang free.

Caloric Deficit – The condition needed to cause weight loss that is created by consuming fewer calories than are burned.

Capillarization – The density of capillary blood vessels in a muscle, particularly the forearms.

Cardiovascular Exercise (Cardio) – Relatively low intensity exercise that stresses the body's systemic aerobic metabolism, especially the heart and lungs' capacity to supply oxygenated blood to the muscles.

Center of Mass (COM) – The point in space around which the climber's body is perfectly balanced.

Climber's Elbow – See Medial or Lateral Epicondylitis.

Closed Crimp – A hand grip position in which all the fingers are bent at ~90 degrees at the PIP joints. The thumb may or may not wrap over the index finger nail. See photo on p55.

Complex Training – A method that combines strength and plyometric exercises to improve speed and power output.

Concentric – A muscle contraction where the muscle shortens.

Contact Strength – The amount of force the fingers can generate during the period of initial contact with a given hold; which is critical to latching a target hold during dynamic moves.

Cross-Bridge Cycling – A process in which the actin and myosin proteins interact and repeatedly pull on each other to contract the Sarcomeres, and muscle fibers.

Cross Sectional Area (CSA) – The sum-total area of contracting muscle fibers in the plane orthogonal to the direction of contraction. A primary factor in muscular force production.

Cytoplasm – Fluid within the cells (particularly the muscle cells). Allows chemicals for muscle contractions to diffuse around the cell to various locations where they are needed.

Dab – When a free-hanging body part incidentally contacts the ground or other "off-limits" supporting structure.

Deadhang – A static, isometric hang from a hangboard (or similar) involving virtually no motion, in which arms are straight and elbows and shoulders are slightly flexed.

Deadpoint – A dynamic move in which the climber catches the target hold at the highest point of flight, with no overshoot, or extraneous motion of the COM. Deadpoints are used to conserve energy, and are required for latching non-positive target holds.

De-Training – The process through which trained muscles revert back to un-trained levels of fitness.

Duration – A quantitative measure of the time required to complete a training activity.

Duty Cycle – The ratio of work time to rest time (a 2 second effort with a 1 second rest, is a duty cycle of 2:1).

Eccentric – A muscle contraction where the muscle lengthens.

Estimated Glycemic Load (eGL)TM – A metric provided by Nutrition Data (<http://nutritiondata.self.com>) that estimates a food's effect on the body's blood sugar level based on its nutrition contents.

Failure (training) – The point at which the body can no longer perform a given exercise.

False Grip – A hand grip in which the hand is wrapped around a protruding hold. See photo on p55.

Fast Twitch Fibers (aka Type IIa and IIX/d or IIb) – Muscle fibers that contract relatively quickly and with great force, but fatigue quickly.

Finger Board – Another term for “Hangboard” (see “Hangboard”).

Flapper – A skin injury in which the skin of a finger pad is cut deeply and a portion of the pad is peeled away.

Flash Pump – A rapidly developing and debilitating pump that typically results from an inadequate warm-up.

Focus – Attention to the task at hand. Essential to effective training and performance.

Fullness FactorTM – A metric provided by Nutrition Data (<http://nutritiondata.self.com>) that estimates a food's satiety based on its nutrition contents.

Functional Hypertrophy – AKA Sarcomere hypertrophy. Growth of the contractile tissues within the muscle, as opposed to non-functional (AKA) sarcoplasmic hypertrophy, which is growth of the energy supply tissues surrounding the muscles.

Gap Analysis – The process of comparing the attributes of a goal climb to the climber's strengths & weaknesses in order to identify areas in need of improvement.

Gaston – A vertical hand grip that lies above, and facing the climber, requiring a thumb-down hand orientation to utilize. See photo in **Chapter Three: Skill Development**, p55.

Goal Route – A specific climb the climber desires to complete. Ideally the climb is selected in advance of training so it can be used to guide training efforts.

Grip Control – The conscious process of relaxing the forearm flexors as much as possible while climbing or resting, in order to delay the onset of fatigue.

Glucose – An energy-dense sugar, derived from carbohydrates that is the preferred fuel for generating ATP to power muscle contractions.

Glycemic Index – The quantitative measure of how quickly blood sugar level rises after consuming a particular food.

Glycemic Load – The cumulative effect of eating a given portion of a particular food that accounts for the food's glycemic index, and its total amount of net carbohydrates.

Glycemic Response – The body's change in blood sugar level as a result of consuming a particular food. This response is measured by the Glycemic Index.

Glycogen – Stored chemical energy that resides in the liver and muscles, and is derived from Glucose through the glycogenesis process. It can be quickly converted to Glucose to support muscle activity.

Glycolysis/Glycolytic Pathway – A complex process for quickly converting glucose and/or glycogen to ATP without oxygen (anaerobically). It eventually leads to accumulating metabolites and muscle fatigue.

Half-Crimp – See Semi-Closed Crimp.

Hangboard – A device used for training finger strength in which athletes perform static hangs from various hand grips while adjusting resistance by adding or removing weight from body weight.

Hangdog – A technique for learning the movement sequences on a route by using the rope for support between learning attempts, as opposed to lowering immediately after a fall.

Hard Bouldering – A training activity in which the climber attempts boulder problems that are too difficult to flash, but can usually be completed in a single session of effort.

Hydrogen Ions (H⁺) – A byproduct of muscle energy metabolism that contributes to fatigue.

Hypertrophy – The volumetric increase of a particular organ or tissue. In strength training, this term is used to describe growth in muscle tissue volume.

Individualization – The deliberate tailoring of a training program to address the specific strengths, weaknesses, and response to training of a given athlete.

Inorganic Phosphate (P_i) – A byproduct of muscle energy metabolism that contributes to fatigue.

Intensity – The level of effort that must be applied to complete a given exercise. For some exercises, intensity is easily quantified in terms of the resistance used. For other exercises, intensity cannot be easily quantified, but it can be qualitatively estimated.

Isolation – The principle of training muscle groups individually.

Isometric – A muscle contraction in which the muscle length remains constant.

Isotonic – A muscle contraction in which the force remains constant while the length of the muscle changes.

APPENDIX

Jib – A tiny artificial foothold that is so small it must be screwed onto the wall (rather than bolted on). See photo on p50.

Ketosis – A condition of low blood sugar level in which stored fat is broken down into fatty acids that can be metabolized in place of glucose to fuel muscle function.

Kick Plate – A small panel covered with footholds that is fastened beneath a hangboard or campus board, allowing mock-climbing. See photo on p162.

Kipping – Swinging the lower body to generate momentum for a campus movement.

Lateral Epicondylitis – (aka Tennis Elbow) A common overuse injury manifesting as acute pain at the point in the elbow where the finger extensor tendons attach to the lateral epicondyle.

Limit Bouldering – Climbing short boulder problems that feature 1 - 2 crux moves (preferably dynamic moves) that are right at the climber's limit.

Limit Route – A route or boulder problem barely within the climber's ability, that requires multiple days of rehearsal and attempts to climb successfully.

Linear Periodization – An approach to periodic training in which various attributes of physical ability are trained in distinct phases or meso-cycles.

Linkage – Successfully climbing sub-sections of a route in one continuous effort during the route rehearsal process.

Linked Bouldering Circuits (LBC) – A high- and mid-intensity form of power-endurance training in which several near-limit boulder problems are linked together to create a circuit of 10 - 40 hand movements requiring 45 seconds to four minutes to climb.

Load – The amount of weight (or resistance) applied to a given exercise.

Logistical Fatigue – Fatigue that develops during multipitch climbs as a result of ancillary, non-climbing tasks.

Macro-Cycle – AKA "Season" A period of time selected by the climber (typically 3 - 4 months) in which periodic training, rest and performance activities are planned.

Maintenance Training – A form of Non-Linear Periodization used to extend the length and quality of a Performance Peak.

Mantel – A hand grip in which downward pressure is placed on a hold with the palm of the hand. See photo on p55.

Maximum Recruitment – A type of physical training designed to encourage the activation of the greatest proportion of motor units for a given contraction.

Maximum Steady State (MSS) – The level of effort that can be sustained by the muscle indefinitely by primarily utilizing aerobic respiration (aka Anaerobic Threshold, Aerobic Threshold, Lactic Threshold).

Medial Epicondylitis – (aka Golfer's Elbow) A common overuse injury manifesting as acute pain at the point in the elbow where the finger flexor tendons attach to the medial epicondyle.

Meso-cycle – AKA "Phase." A period of time within a Macro-Cycle (typically 2 - 4 weeks) focused on a particular type of training (such as Base-Fitness, Strength, Power, or Power-Endurance).

Metabolites – Byproducts of muscle energy metabolism that interfere with muscle function and cause fatigue (including H^+ , P_i , NADH, ADP).

Mitochondria – An organelle found in many cells that aerobically supplies energy (ATP) for cell function (especially muscles).

Mono – A pocket hold that admits only a single finger.

Motor Neuron – Nerve cells that carry signals from the Central Nervous System to the muscles to initiate muscle contractions.

Motor Unit – The smallest functional unit of the neuromuscular system, composed of bundles of one or more muscle fibers, and controlled by a single Motor Neuron.

Muscle Fiber – A long tubular cell composed of numerous myofibrils.

Myofibrils – The smallest individual contractile units in the muscle.

Myosin – Thick filament proteins within the muscle's Sarcomeres that bind to Actin proteins to contract the muscle.

Near-Limit Route/Redpoint – A route or boulder problem that is one or two letter grades easier than a Limit Route, and can typically be climbed successfully in two or three days of effort.

Net Carbs – The portion of a food's total carbohydrates that will be absorbed by the body, calculated as the number of grams of total carbs minus grams of dietary fiber.

Non-Linear Periodization (NLP) – An approach to periodic training in which various attributes of physical ability are trained concurrently, in a single training session or in week-long micro-cycles.

Non-Steroidal Anti Inflammatory Drugs (NSAIDs) – Drugs commonly used for temporary pain relief that are known to delay healing of soft tissue.

Occlusion – Reduced flow of blood to the muscles during muscle contractions.

Open Grip – A hand grip in which the PIP joint is only slightly flexed. See photo on p55.

Outdoor Mileage – A form of base-fitness training, much like ARC training, in which a high volume of climbing (typically 6 - 12 pitches in a day) is performed at a moderate intensity (approximately two number-grades below onsight level).

Overload – Training stress must exceed the body's baseline capability in order to stimulate physical adaptation.

Overuse Injury – An injury that develops gradually over time from excessive stress, as opposed to an Acute Injury.

Oxydative Phosphorylation – An aerobic process that occurs within the mitochondria for converting the metabolites NADH, H⁺ and P_i, along with oxygen, into ATP.

Pace – The rate at which a sequence is climbed. Pace can vary within a given ascent, i.e., climbing faster or slower.

Performance – The act of climbing on rock or during competition, for the purpose of achieving a goal.

Performance Peak – The period during which the athlete expects to experience optimal performance. The length and timing of the Performance Peak is determined by the timing and effectiveness of the training cycle, and can be difficult to precisely control.

Performance Phase – The scheduled mesocycle within a macro-cycle in which the athlete attempts to perform outdoors on rock as much as possible. The length and timing of the Performance Phase is determined by the athlete.

Periodization – A strategy for physical training in which different training activities are performed in accordance with a carefully designed schedule to achieve a synergistic effect on overall performance that results in a *performance peak* at a predictable time.

Phase – A period of time (typically 2 - 4 weeks) within a periodic training program focused on a particular type of physical training.

Pinch – A hand grip in which the thumb is used. See photo on p55.

Phosphagen Pathway – The fasted method for ATP production, used by the muscles for immediate action, and in the first few seconds of muscle activity. Uses the creatine kinase reaction to break creatine phosphate into ATP.

Plyometrics – Dynamic exercises that require the muscles to exert maximum force quickly in order to increase speed and power.

Proximal Interphalangeal (PIP) Joint – The knuckle joint between the two finger bones of each finger that are nearest the palm of the hand.

Plateau – Stagnation in performance caused by the Law of Diminishing Returns.

Power – The ability to quickly exert high force on a set of holds.

Power-Endurance – The ability to perform multiple near-maximal climbing moves without rest (typically in bouts lasting 30 - 180 seconds). During power-endurance climbing, the intensity of the moves is sufficiently high that recruitment of fast-twitch muscle fibers is required.

Power Spot – A spotting technique in which the spotter places his hands on the climber's hips and/or low back and pushes gently to remove weight from the climber to assist in learning difficult moves.

Practice Falls – Progressively longer falls used to condition the lead climber to exposure and reduce falling anxiety.

Project – A goal route that requires more than one attempt to send.

Progressive Overload – The systematic and iterative application of incremental training stress and subsequent recovery to gradually improve the body's capabilities over the previous baseline.

Progression – Improvement over time.

Pulleys (finger) – The ligaments that hold the finger flexor tendons in place to facilitate finger joint movement.

Quantification – The process of measuring training intensity and duration in order to document and control volume and progression.

Range of Motion – The distance a joint can freely move.

Reactive Training – A method of shock-loading the muscles in order to activate their inherent stretch-shortening injury-defense reflex and maximize muscle fiber recruitment.

Recovery – A period of rest following physical training during which supercompensation occurs.

Recruitment – Describes the proportion of active motor units contributing to a given muscular contraction.

Regularity – Training must be performed repeatedly at reasonable time intervals in order to avoid de-training.

Rehabilitation – The process of recovering injured tissue to a healthy state through methodical and controlled application of stress and rest.

Repetition – A term used in physical training to denote a single exercise cycle, such as lifting and lowering a weight.

Resistance – The quantity of weight (or load) applied to a given exercise.

Reversibility – Physical adaptations achieved through training can be lost just as easily as they are gained, if training stimulus is not maintained or increased.

Robust – Able to perform well under various (often poor) conditions.

Roped Bouldering – Working individual crux moves on a difficult route, in preparation for an eventual redpoint. The intensity, duration of effort, and duty cycle are similar to that of bouldering, but the climber is *hangdogging* on a rope to allow access to the most critical moves.

Route Intervals (aka laps, or lapping) – A lower-intensity power-endurance training exercise in which a pump route is climbed several times in succession, with a specified rest period between laps.

Route Pyramid – A visual representation of a climber's "resume" of single pitch rock climbing ascents; an approach to climbing progression that emphasizes building a broad base of experience on routes that are below the climber's maximum ability.

Sarcomeres – The segmented units of the muscle's myofibrils that shorten in a contraction.

Sarcoplasmic Reticulum (SR) – An organelle within the muscle cells that stores and provides calcium ions for muscle contractions.

Satiety – The feeling of satisfaction, or fullness from eating.

Season – AKA "Macro-Cycle" A period of time selected by the climber (typically 3 - 4 months) in which training, rest and performance activities are planned.

Self-Talk – The internal dialogue a climber engages in while climbing. Typically, *positive* self-talk is most beneficial.

Semi-Closed Crimp – AKA Half Crimp, a hand grip position in which the Index and Pinkie fingers are straight, and the Middle & Ring fingers are bent ~90 degrees at the PIP joint. See photo on p55.

Set – A term used in physical training to denote a group of repetitions performed consecutively, such as lifting and lowering a weight five times in quick succession.

Sidepull – A vertically-oriented hand grip that faces away from the climber. See photo on p55.

Size Principle – During muscular contraction, smaller, slow twitch motor units tend to be activated first, and the larger and stronger fast twitch units tend to be activated last.

Skill Development – The process of acquiring new technical movement abilities, refining them through practice, and finally stress-proofing those abilities for use during performance.

Slow Twitch Fibers (aka Type I) – Muscle fibers that contract relatively slowly and with less force, but are able to sustain contractions for a longer duration.

Specificity – An athlete will primarily achieve physical adaptations in the systems that are stimulated by the training activity.

Stamina – The body's ability to endure multiple difficult climbing efforts, or pitches, throughout a climbing day.

Strength (physical strength) – The body's ability to exert high force on a climbing hold.

Strength (performance strength) – A skill or ability required for climbing that the climber performs well (relative to other skills and abilities).

Stress-Proofing – A progressive process for improving the ability to perform a new skill in high-pressure situations.

Super-Compensation – The body's adaptive response to physical training stress.

Supinate – Outward rotation of a limb. For example, an arm rotating from a palms-down position to palms-up.

Systemic – A condition that affects the entire body as a whole.

Tapering – Steadily reducing the total volume of exercise near the end of a season in order to augment a performance peak.

Time Under Tension (TUT) – The duration of a training activity spent under load (total exercise time, minus the duration of any rest intervals).

Time-Value of Climbing Ability – Time invested today in training, will pay off with more sending power in the future.

Transfer – Train in a manner that replicates the conditions of performance to the extent possible.

Training – An approach to improve climbing performance that is systematic, disciplined, and science-based.

Training Stress – The cumulative fatigue resulting from a climbing workout or performance day.

Treadwall® – An artificial climbing wall that rotates as the climber climbs, like a vertically-oriented treadmill.

Tweaky – A hold or move that qualitatively feels likely to cause injury.

Twitch – A mechanical response in the motor unit caused by an action potential from a motor neuron.

Undercling – A downward-oriented hand grip, on which the climber must apply upward pressure. See photo on p55.

Variation – Adjustments to training methods and stress to avoid performance plateaus.

Volume – The product of training intensity and training duration.

Warmup Boulder Ladder – A warmup activity in which the climber completes a series of boulder problems beginning with V-Easy, and progressing up to the climber's flash limit.

Weakness – A skill or ability required for climbing that the climber performs poorly (relative to other skills and abilities).

Workload – The total volume of training stress applied to an athlete during a given time period.

Wunderkind – A phenomenal climber who began climbing regularly as a child.

Zone – The mental state of focus and concentration required for optimal performance.